

TUTORIAL SHEET

Q1. Convert the following:

(i) $(3287.5100098)_{10} = (?)_2 = (?)_8$

(ii) $(675.625)_{10} = (?)_2 = (?)_{16}$

Q2. Perform subtraction using **r's** and **(r-1)'s** complement method :

(i)

$$\begin{array}{r} (587C)_H \\ -(974B)_H \\ \hline (?)_{10} \end{array}$$

(ii)

$$\begin{array}{r} (0111)_2 \\ -(1101)_2 \\ \hline (?)_8 \end{array}$$

(iii)

$$\begin{array}{r} (564)_8 \\ -(7C3)_H \\ \hline (?)_2 \end{array}$$

Q3. The base of the number system for the addition operation $24+14=41$ to be true is.....

Q4. (i) A particular number system having base B is given as $(\sqrt{41})_B = (5)_{10}$. The value of B is

(ii) Given that $(E0B)_H - (ABF)_H = Y$. The Radix 8's complement of Y . is.....

Q5.

(i) The state of a 12-bit register is 100010010111. What is its content if it represents :

(a) three decimal digits in BCD

(b) three decimal digits in Excess-3 code

(ii) Convert the following :

(a) $(10010)_2$ to gray code.

(b) $(10111)_{\text{Gray}}$ to binary code